Optimal Bitcoin Trading with Inverse Futures

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Abstract

We consider an optimal trading problem for an investor who trades Bitcoin spot and Bitcoin inverse futures, plus a risk-free asset. The investor seeks an optimal strategy to maximize her expected utility of terminal wealth. We obtain explicit solutions to the investor's optimal strategies under both exponential and power utility functions. Empirical studies confirm that optimal strategies perform well in terms of Sharpe ratio and Sortino ratio and beat the long-only strategy in Bitcoin spot.

Key words: Bitcoin; Inverse Futures; Optimal Investment; Utility Maximization

1 Introduction

We study optimal trading problems for an investor who trades in a financial market with one risk-free asset, Bitcoin spot and one Bitcoin *inverse* futures contract. We find optimal trading strategies in the Bitcoin spot and futures in closed forms when the investor seeks to maximize her expected exponential or power utility of terminal wealth. Economic analysis shows that such optimal strategies perform reasonably well in terms of Sharpe ratio and Sortino ratio.

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1.1 Bitcoin Futures

Bitcoin (BTC) is a cryptocurrency created by Satoshi Nakamoto in 2008 (see Nakamoto (2008)), and has the largest market capitalization of over 203 billion US dollars (USD, with symbol $) among all cryptocurrencies (See Table A.1). There is an ongoing debate on whether Bitcoin should be regarded as a real currency (see Yermack (2015)). Nevertheless, Bitcoin is among the most traded assets in recent years,[1] and is of great interest to both the industry and academia. Please refer to Narayanan et al. (2016) for a comprehensive introduction on Bitcoin.

The year of 2017 witnessed the biggest increase on the Bitcoin price, rising from as low as $800 in January to the staggering highest at $19,783.06 on December 17, converting to an unprecedented annual return of more than 2,300%. Given the strong market interest and demand on hedging and speculation, the Chicago Board Options Exchange (CBOE) introduced the Bitcoin futures contract and opened trading on December 10, 2017. The Chicago Mercantile Exchange (CME) soon listed its Bitcoin futures for trading on December 18, 2017. The Bitcoin futures contracts of regulated exchanges CBOE and CME are *standard* contracts, both denominated in USD; see Table A.3 for more information on these contracts. More importantly, the CBOE and CME Bitcoin futures are settled in USD, which contradicts the essential purpose of Bitcoin as a peer-to-peer and decentralized electronic settlement system.

In response, more and more Bitcoin exchanges (less regulated comparing to CBOE and CME), such as BitMEX, Binance and Kraken, have introduced *inverse* futures contracts,[2] which have a face value (contract size) of certain USD and are settled in BTC, exactly the opposite to those standard contracts of CBOE and CME. In other words, an inverse futures contract treats USD as the underlying "commodity" and uses BTC as the currency for settlement, i.e., USD/BTC pair instead of BTC/USD pair. Let K be the face value of an inverse futures contract and F the reference futures price, both quoted in USD, then the equivalent value of the futures contract is K/F number of Bitcoins, with an *inverse* relation to the futures price F. Please refer to Table A.4

[1] Table A.1 records a daily trading volume equivalent to nearly 7 billion US dollars on a give day.

[2] To the best of our knowledge, Bragin (2015), co-founders of ICBIT, first proposed the Bitcoin inverse futures and the Russian ICBIT trading platform was the very first Bitcoin exchange to offer Bitcoin futures contracts, later acquired by Swedish-based Bitcoin exchange Safello. See news report on https://bravenewcoin.com/insights/safello-acquires-icbit-and-appoints-founder-as-new-cto.

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for information on the Bitcoin futures contracts traded on BitMEX. Let us use a toy example to illustrate how the profit and loss (P&L) of an inverse futures contract is calculated.

Example 1.1. *Consider an investor who trades a Bitcoin inverse futures contract, which has* $K = \$1$ *per contract size. For simplicity, we ignore the margins on the futures contracts. Suppose the investor enters into one long position in the futures at* t_1 *when* $F_{t_1} = \$100$ *and closes the position at* t_2 *when* $F_{t_2} = \$200$*. The trading profit is then given by*

$$\frac{K}{F_{t_1}} - \frac{K}{F_{t_2}} = \frac{1}{100} - \frac{1}{200} = 0.005 \ BTC.$$

If the spot price of Bitcoin at t_2 *is \$205, then the investor can either keep the profit in BTC or sells 0.005 BTC in the spot market to pocket \$1.025.*

Bitcoin inverse futures is dominating the markets and the top five exchanges by futures trading volumes are BitMEX, bitFlyer, Deribit, CoinFlex, and CryptoFacilities. BitMEX has the largest 24h trading volume of 2.74 \$bn, followed by bitFlyer with only 0.97 \$bn (See Table A.2). Due to disappointing market shares, CBOE discontinued its Bitcoin futures contracts after June 19, 2019, leaving CME the only venue for trading standard Bitcoin futures contracts. In the meantime, even the most liquid Bitcoin futures contract at CME has a miserable trading volume.[3] Recently, Bakkt announces that it will be launching its Bitcoin futures in the U.S. on September 23, 2019. The Bitcoin futures on Bakkt will be exchange-traded on the Intercontinental Exchange (ICE) Futures U.S. and cleared by ICE Clear U.S., which are federally regulated by the Commodity Futures Trading Commission (CFTC).[4]

1.2 Literature Review and Summary

Bitcoin, or cryptocurrency, is among the hottest areas in economics and finance over recent years, and the related literature is growing at a rapid scale into various directions. Given the length of the paper, we only give a brief review on works that focus on Bitcoin futures. The first line of research

[3] See volume information on https://www.cmegroup.com/trading/equity-index/us-index/bitcoin.html.
[4] See reports on https://www.bakkt.com/index.

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investigates the price discovery of Bitcoin spot and futures. Both Corbet et al. (2018) and Baur and Dimpfl (2019) find that the price discovery is dominantly driven by the Bitcoin spot market. But Kapar and Olmo (2019) arrive at the exactly opposite conclusion, i.e., the Bitcoin futures market dominates the price discovery. The second line of research concerns the impact of Bitcoin futures on the spot prices. Hale et al. (2018) argue that the inception of Bitcoin futures in December 2017 contributed to the subsequent fall of Bitcoin spot price. Köchling et al. (2018) claim that Bitcoin futures helps improving the market efficiency of Bitcoin. The third line of research investigates the use of Bitcoin futures in risk management (mainly as a hedging tool). Corbet et al. (2018) argue that using Bitcoin futures in hedging may increase risk, but Sebastião and Godinho (2019) find Bitcoin futures to be an effective hedging tool for not only Bitcoin but other cryptocurrencies.

The optimal trading problem studied in this paper belongs to the family of optimal investment problems in finance, which are well studied in the literature; see the seminal works of Merton (1969, 1971) (hence often referred to as *Merton's* problem). Duffie and Jackson (1990) is among the earliest to study the optimal hedging problem with standard futures. For further development along this line, please see Duffie and Richardson (1991), Dai et al. (2011), Leung et al. (2016), Angoshtari and Leung (2019) and the references therein. For Merton's problem with derivatives (not necessarily futures), we refer to Carr et al. (2001) for an exponential Lévy price model and Liu and Pan (2003) for a jump-diffusion model with stochastic volatility. Recent generalizations include Yan and Li (2008) for incorporating stochastic exchange rate and Escobar et al. (2015) for considering ambiguity aversion in the analysis.

The contributions of this paper are threefold. First, we consider an optimal trading problem with both Bitcoin spot and inverse futures, which to our best knowledge has not been studied before.[5] Despite being popular derivatives contracts traded across global markets, Bitcoin inverse futures rarely attracts the spot light in academia. In fact, the majority of the literature on Bitcoin futures (including those reviewed above) is devoted to the studies of CBOE/CME standard contracts, which are illiquid and have much less market shares. Second, we solve the investor's optimal trading problem using two different methods and obtain optimal trading strategies in closed forms

[5]The only exception to our awareness is Deng et al. (2019), which studies an optimal hedging problem with Bitcoin inverse functions under the mean-variance criterion.

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when the investor wants to maximize her expected exponential or power utility of terminal wealth. Third, through empirical studies, we find that the optimal trading strategies yield good Sharpe ratio and Sortino ratio and outperform the long-only strategy in Bitcoin spot.

The rest of the paper is organized as follows. In Section 2, we introduce a continuous-time financial market and formulate an optimal trading problem with Bitcoin inverse futures. In Section 3, we outline two fundamental methods to solve stochastic control problems. We then apply these two methods to solve the optimal trading problem under exponential and power utility functions in Section 4. In Section 5, we conduct empirical studies to investigate the performance of optimal trading strategies. We summarize our concluding remarks in Section 6. Finally, we place tables regarding Bitcoin and futures contracts, along with technical derivations and proofs, in Appendices A-C.

2 Problem Setup

Let us fix a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ over a finite time horizon $[0, T]$, with $T > 0$. A standard n-dimensional Brownian motion $W = (W_1, W_2, \ldots, W_n)^\top$ is defined on this space, where $n \geq 2$, W_i and W_j are independent for all $i \neq j$ and $\cdot^\top$ denotes the usual transpose operation. We take the filtration $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}$ to be the augmented filtration of W and set $\mathcal{F}_0 = \{\emptyset, \Omega\}$. In a continuous-time financial market, we consider a representative investor who trades *three* assets: a risk-free asset, Bitcoin spot currency, and a Bitcoin *inverse* futures contract. Here, the inverse futures contract either has an expiry date $T' \geq T$ or is a perpetual contract.[6] We assume that the risk-free interest rate is zero, i.e., we take the risk-free asset as the numéraire and normalize it to 1. Denote by $S = (S_t)_{t \geq 0}$ and $F = (F_t)_{t \geq 0}$ the Bitcoin spot price and futures *reference* price, respectively. Note that both the spot price S and the futures price F are quoted in the fiat currency (USD). Following Duffie and Jackson (1990), we model the Bitcoin spot price S and futures price

[6] Similar to traditional futures contract, perpetual futures contract is a derivative product, but does not have an expiry or settlement date. More details are available on https://www.bitmex.com/app/perpetualContractsGuide.

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F by

$$\mathrm{d}S_t = S_t\left(\mu_1\,\mathrm{d}t + \sum_{i=1}^{n}\sigma_{1,i}\,\mathrm{d}W_{i,t}\right) = S_t\big(\mu_1\,\mathrm{d}t + \sigma_1^\top\,\mathrm{d}W_t\big), \quad S_0 > 0, \tag{2.1}$$

$$\mathrm{d}F_t = F_t\left(\mu_2\,\mathrm{d}t + \sum_{i=1}^{n}\sigma_{2,i}\,\mathrm{d}W_{i,t}\right) = F_t\big(\mu_2\,\mathrm{d}t + \sigma_2^\top\,\mathrm{d}W_t\big), \quad F_0 > 0, \tag{2.2}$$

where μ_i and $\sigma_{i,j}$ are constant and $\sigma_i := (\sigma_{i,1}, \sigma_{i,2}, \ldots, \sigma_{i,n})^\top$ for all $i = 1, 2$ and $j = 1, \ldots, n$. Let $\sigma^\top := (\sigma_1,\ \sigma_2)$ and assume $\sigma\sigma^\top$ is non-singular, i.e.,

$$\det(\sigma\sigma^\top) = (\sigma_1^\top\sigma_1) \times (\sigma_2^\top\sigma_2) - (\sigma_1^\top\sigma_2)^2 = \nu_1\nu_2 - \nu_3^2 \neq 0, \tag{2.3}$$

where we define constants ν_i by

$$\nu_i := \sigma_i^\top\sigma_i = \sum_{j=1}^{n}\sigma_{i,j}^2, \quad i = 1, 2, \quad \text{and} \quad \nu_3 := \sigma_1^\top\sigma_2 = \sum_{j=1}^{n}\sigma_{1,j}\sigma_{2,j}. \tag{2.4}$$

Condition (2.3) is assumed throughout the paper.

Remark 2.1. *Linear algebra gives us*

$$\det(\sigma\sigma^\top) = \nu_1\nu_2 - \nu_3^2 = \sum_{i\neq j}(\sigma_{1i}\sigma_{2j} - \sigma_{1j}\sigma_{2i})^2 \geq 0,$$

as matrix $\sigma\sigma^\top$ is non-negative definite. Condition (2.3) is equivalent to assume $\sigma\sigma^\top$ is positive definite, and hence, is invertible. $\nu_1, \nu_2 > 0$ is obvious. In the markets, S and F are positively correlated, thus we anticipate $\nu_3 > 0$.

As discussed in the introduction, an inverse Bitcoin futures contract has a size (face value) fixed to K unit of the fiat currency (USD), and is settled in BTC. For instance, one Bitcoin futures contract traded on BitMEX (`www.bitmex.com`) is fixed to \$1 USD worth of Bitcoins (i.e., $K = 1$). With a fixed face value K (in USD), the number of Bitcoins per contract is given by K/F_t at time t. Let $\widehat{F}$ denote the value of a long position in one inverse contract, quoted in BTC. We obtain the

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value change of the inverse contract $\Delta\widehat{F}$ over $(t, t+\Delta t)$ by

$$\Delta\widehat{F}_t = \frac{K}{F_t} - \frac{K}{F_{t+\Delta t}} = \frac{K}{F_t}\frac{\Delta F_t}{F_t + \Delta F_t} = \frac{K}{F_t}\frac{\Delta F_t}{F_t}\frac{1}{1+\Delta F_t/F_t} = \frac{K}{F_t}\frac{\Delta F_t}{F_t}\sum_{i=0}^{\infty}\left(-\frac{\Delta F_t}{F_t}\right)^i.$$

By letting $\Delta t \to 0$ and using (2.2), we get the dynamics of $\widehat{F}$ by

$$\mathrm{d}\widehat{F}_t = \frac{K}{F_t}\big((\mu_2 - \nu_2)\,\mathrm{d}t + \sigma_2^\top \mathrm{d}W_t\big). \tag{2.5}$$

Note that $\Delta\widehat{F}$ (or $\mathrm{d}\widehat{F}$) is expressed in units of Bitcoin (BTC), not the fiat currency (USD). Under the mark to market mechanism, the realized profit & loss (P&L) in the fiat currency (USD) at time t per futures contract is then given by $S_t\,\mathrm{d}\widehat{F}_t$. Let us introduce the ratio process $Z = (Z_t)_{0\le t\le T}$, where $Z_t := S_t/F_t$. Using Itô's formula, we have

$$\mathrm{d}Z_t = Z_t\big(\mu_1 - \mu_2 + \nu_2 - \nu_3\big)\,\mathrm{d}t + Z_t\big(\sigma_1 - \sigma_2\big)^\top \mathrm{d}W_t. \tag{2.6}$$

The investor's trading strategy is self-financing and consists of a pair of processes, denoted by $u = (u_t)_{0\le t\le T} := (\theta, N) = (\theta_t, N_t)_{0\le t\le T}$. Here, θ_t is the dollar *amount* of wealth invested in the Bitcoin spot currency and N_t is the *number* of the Bitcoin inverse futures contracts held at time t. $N > 0$ (resp. $N < 0$) denotes a long (resp. short) position in the futures contract. Recall that, with futures position N, the corresponding P&L in USD is given by $NS\,\mathrm{d}\widehat{F}$. Denote by $X^u = (X^u_t)_{0\le t\le T}$ the associated wealth process of the investor, if she follows a trading strategy u. We write X to replace X^u, if there is no confusion, and only use X^u if we want to emphasize the dependence on control u. By the above definition, we have

$$X_t = x + \int_0^t \theta_s \frac{\mathrm{d}S_s}{S_s} + \int_0^t N_s S_s\,\mathrm{d}\widehat{F}_s, \qquad \forall\, t \in [0,T],$$

where $X_0 = x > 0$ is the investor's initial wealth. Using (2.1), (2.5) and (2.6), we obtain

$$\mathrm{d}X_t = \big(\mu_1\theta_t + K(\mu_2 - \nu_2)Z_tN_t\big)\,\mathrm{d}t + \big(\theta_t\sigma_1 + KZ_tN_t\sigma_2\big)^\top\,\mathrm{d}W_t, \tag{2.7}$$

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where ν_2 is defined by (2.4).

Let us define space $\mathbb{L}^2[t,T]$, for all $0 \leq t < T$, by

$$\mathbb{L}^2[t,T] := \left\{ \omega = (\omega_s)_{t \leq s \leq T} \, : \, \omega \text{ is progressively measurable and } \mathbb{E}_t \left[\int_t^T \omega_s^2 \, \mathrm{d}s \right] < \infty \right\}, \quad (2.8)$$

where $\mathbb{E}_t$ denotes conditional expectation operator given $\mathcal{F}_t$. Denote by $\mathcal{A}(t,x,z)$ the *admissible* set of trading strategies over period $[t,T]$ with initial conditions $X_t = x$ and $Z_t = z$. At $t = 0$, we write the admissible set by $\mathcal{A}(x)$, since $Z_0 = S_0/F_0$ is known. We define the admissible set as follows (see, e.g., Duffie and Jackson (1990) and Angoshtari and Leung (2019)).

Definition 2.2. *A trading strategy u is called an admissible strategy (i.e., $u \in \mathcal{A}(t,x,z)$) if (i) θ, $ZN \in \mathbb{L}^2[t,T]$, where $\mathbb{L}^2[t,T]$ is defined by (2.8), and (ii) the SDE of X in (2.7) has a unique strong solution.*

Let U be a standard utility function, namely, U is strictly increasing and strictly concave over its domain. We define the objective functional $J(x;u)$ for any $u \in \mathcal{A}(x)$ by

$$J(x;u) := \mathbb{E}\big[U(X_T)\big],$$

where $\mathbb{E}$ is the expectation operator under $\mathbb{P}$ (given $X_0 = x > 0$).

The investor seeks an optimal trading strategy, denoted by u^*, to maximize the expected utility of the terminal wealth over the set of all admissible strategies:

$$V(x) = \sup_{u \in \mathcal{A}(x)} \mathbb{E}\big[U(X_T)\big]. \quad (2.9)$$

We call V the investor's *value function* and u^* the investor's optimal trading strategy or optimal control.

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3 Methodology

In this section, we outline two methods for solving Problem (2.9): the Hamilton-Jacobi-Bellman (HJB) method in Section 3.1 and the martingale method in Section 3.2. We postpone technical details in Appendix B.

3.1 The HJB Method

The first method we apply is the famous HJB method, which is based on the dynamic programming principle (DPP). Please see Yong and Zhou (1999) and Fleming and Soner (2006) for general theory.

Let $f = f(t,x,z) : [0,T] \times \mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}$ and $f \in \mathrm{C}^{1,2,2}$. Denote by $f_{\cdot}$ the partial derivative of function f with respect to (w.r.t) the corresponding argument, e.g., $f_x = \frac{\partial f}{\partial x}$. For any $u \in \mathcal{A}(t,x,z)$ and $f \in C^{1,2,2}$, define the following operators

$$\begin{aligned}
\mathcal{L}_1 f(t,x,z) &:= (\mu_1 - \mu_2 + \nu_2 - \nu_3)\, z \cdot f_z + \frac{1}{2}(\sigma_1 - \sigma_2)^\top (\sigma_1 - \sigma_2) z^2 \cdot f_{zz}, \\
\mathcal{L}_2^u f(t,x,z) &:= (\mu_1 \theta + K(\mu_2 - \nu_2) z N)\ f_x + \frac{1}{2}\left(\theta\sigma_1 + KzN\sigma_2\right)^\top \left(\theta\sigma_1 + KzN\sigma_2\right)\ f_{xx} \\
&\quad + \left(\sigma_1 - \sigma_2\right)^\top \left(\theta\sigma_1 + KzN\sigma_2\right)\ z \cdot f_{xz}.
\end{aligned}$$

To use the DPP to derive the HJB, we consider the dynamic version of Problem (2.9):

$$V(t,x,z) = \sup_{u \in \mathcal{A}(t,x,z)} \mathbb{E}_{t,x,z}\left[U(X_T)\right], \tag{3.1}$$

where $\mathbb{E}_{t,x,z}[\cdot] = \mathbb{E}[\cdot | X_t^u = x, Z_t = z]$. As standard in stochastic control theory, we provide a verification theorem to Problem (3.1) in a general setting. We refer readers to Section 4.3 of Yong and Zhou (1999) and Section III.8 of Fleming and Soner (2006) for related proofs.

Theorem 3.1. *Let $v = v(t,x,z) : [0,T] \times \mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}$ and $v \in \mathrm{C}^{1,2,2}$. Suppose the function v satisfies the Hamilton-Jacobi-Bellman (HJB) equation*

$$v_t + \mathcal{L}_1 v + \sup_{u \in \mathcal{A}(t,x,z)} \mathcal{L}_2^u v = 0, \qquad v(T,x,z) = U(x). \tag{3.2}$$

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Then $v(t,x,z) = V(t,x,z)$ (resp. $v(0,x,Z_0) = V(x)$) is the value function to Problem (3.1) (resp. Problem (2.9)). If strategy $u^* = (u_s^*)_{t\le s\le T}$, defined by

$$u_s^* = \arg\max_u \ \mathcal{L}_2^u\, v(s, X_s^*, Z_s), \qquad \forall\, 0 \le t \le s \le T, \tag{3.3}$$

is admissible, then u^* is an optimal trading strategy to Problem (3.1). Here X^* is the solution to (2.7) under strategy u^*.

In general, there does not exist explicit solutions of the HJB equation (3.2) jointly with the optimization problem (3.3). We are able to obtain u^* explicitly when the utility function takes certain forms (see the main results in the next section).

3.2 The Martingale Method

Next, we discuss how to use the martingale method to solve Problem (2.9). For works on the development of the martingale method, see for instance Cox and Huang (1989) and the monograph Karatzas and Shreve (1998). What we present in the sequel is based on Zou and Cadenillas (2014). We begin with a lemma that provides a sufficient condition for an optimal control to Problem (2.9).

Lemma 3.2. *Suppose there exists an admissible control u^* (i.e., $u^* \in \mathcal{A}(x)$) such that*

$$\mathbb{E}\left[U'(X_T^*)\, X_T^u\right] = \ \text{constant}, \qquad \forall\, u \in \mathcal{A}(x), \tag{3.4}$$

then u^* is an optimal trading strategy (an optimal control) to Problem (2.9). Here X_T^* is the terminal wealth under control u^*.

Proof. The assertion follows from the Taylor expansion

$$\mathbb{E}[U(X_T^u)] = \mathbb{E}[U(X_T^*)] + \mathbb{E}\left[U'(X_T^*)\,(X_T^u - X_T^*)\right] + \frac{1}{2}\mathbb{E}\left[U''(X_T^*)\,\xi^u\right],$$

where ξ^u is between X_T^u and X_T^*, and the fact that $U'' < 0$. □

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The essential idea is to introduce a new probability measure $\mathbb{Q}$ by

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \frac{U'(X_T^*)}{\mathbb{E}\left[U'(X_T^*)\right]} := H_T$$

and notice that the optimality condition (3.4) reduces to $\mathbb{E}^{\mathbb{Q}}[X_T^u]$ is a constant for *all* admissible controls. We then pick two special admissible controls $u_1 = (\theta_1, N_1)$ and $u_2 = (\theta_2, N_2)$ to arrive at two equations for θ^* and N^*, where

$$\theta_{1,t} = x \cdot \mathbb{I}_{t \le \tau}, \quad N_{1,t} = 0, \qquad \text{and} \qquad \theta_{2,t} = 0, \quad N_{2,t} = \frac{1}{K\, Z_t} \cdot \mathbb{I}_{t \le \tau},$$

for any stopping time $\tau \in [0, T]$. We then solve these two equations to get an optimal control.

4 Explicit Solutions for Exponential and Power Utility Functions

We apply the two methods introduced in the previous section to obtain closed form solutions to Problem (2.9) for an exponential utility in Section 4.1 and a power utility in Section 4.2. Both exponential and power utility functions belong to the family of hyperbolic absolute risk aversion (HARA) utility, and are well adopted in the portfolio optimization literature (see Merton (1969, 1971)).

4.1 The Exponential Utility Case

In this subsection, we consider an exponential utility function U given by

$$U(x) = -\frac{1}{\gamma}\, e^{-\gamma x}, \qquad \gamma > 0, \tag{4.1}$$

and present the key result in Theorem 4.1.

Theorem 4.1. *Suppose U is given by (4.1). The optimal trading strategy $u^* = (\theta^*, N^*)$ to Problem*

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(2.9) *is given by*

$$\theta_t^* = \frac{\mathcal{C}_1}{\gamma} \qquad \textit{and} \qquad N_t^* = \frac{\mathcal{C}_2}{\gamma K Z_t}, \qquad \forall\, t \in [0,T], \tag{4.2}$$

where constants $\mathcal{C}_1$ *and* $\mathcal{C}_2$ *are defined by*

$$\mathcal{C}_1 := \frac{\mu_1\nu_2 - \mu_2\nu_3 + \nu_2\nu_3}{\nu_1\nu_2 - \nu_3^2} \qquad \textit{and} \qquad \mathcal{C}_2 := \frac{\mu_2\nu_1 - \mu_1\nu_3 - \nu_1\nu_2}{\nu_1\nu_2 - \nu_3^2}. \tag{4.3}$$

Proof. We provide two proofs in Appendix C. □

Let us interpret KN as the dollar *amount* "invested" in the futures and recall K is the size per futures contract in dollar amount. The optimal trading strategy u^* in (4.2) suggests that optimal dollar amount invested in both spot and futures assets are inversely proportional to the *absolute risk aversion* γ. Namely, a more risk averse investor will reduce her holdings in both Bitcoin spot and futures and allocate larger amount to the risk-free asset. We easily see $\frac{\partial \mathcal{C}_1}{\partial \mu_1} > 0$, and if $\nu_3 > 0$ as explained in Remark 2.1, $\frac{\partial \mathcal{C}_1}{\partial \mu_2} < 0$. Therefore, as the appreciation rate of Bitcoin spot (resp. futures) increases, the investor spends more (resp. less) amount in Bitcoin spot; the contrary qualitative conclusion holds true on the side of futures. We reorganize KN_t^* into $KN_t^* = \frac{\mathcal{C}_2}{\gamma}\frac{1}{Z_t} = \frac{\mathcal{C}_2}{\gamma}\frac{F_t}{S_t}$, and suppose $\mathcal{C}_2 < 0$ in the following analysis. If $\Delta Z > 0$ (resp. $\Delta Z < 0$), the investor should short less (resp. more) futures contracts. Notice that $\Delta Z > 0 \Leftrightarrow \frac{\Delta S}{S} > \frac{\Delta F}{F}$, where $\frac{\Delta S}{S}$ and $\frac{\Delta F}{F}$ are the return on Bitcoin spot and the nominal return on Bitcoin reference futures, respectively. However, the impact of volatilities (σ_i, $i = 1, 2$) on θ^* and N^* is unclear, at least from the analytical point of view.

Remark 4.2. *Using* (2.3) *and* (2.4)*, we have*

$$\begin{pmatrix} \mathcal{C}_1 \\ \mathcal{C}_2 \end{pmatrix} = \left(\sigma\sigma^\top\right)^{-1} \begin{pmatrix} \mu_1 \\ \mu_2 - \nu_2 \end{pmatrix} := \pi_M, \quad \textit{where } \sigma\sigma^\top = \begin{pmatrix} \nu_1 & \nu_3 \\ \nu_3 & \nu_2 \end{pmatrix}. \tag{4.4}$$

In (4.4)*, we follow the convention and denote the r.h.s. by* π_M*, the so called* Merton ratio*, although in the context of exponential utility maximization, it is optimal* amount *not proportion invested in*

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the risky assets that is a fixed constant (π_M/γ). *An important implication of* (4.4) *is that the optimal trading strategy* u^* *under our setup is the same as the one in a financial market with the inverse contracts replaced by asset* $\tilde{S}$, *whose dynamics follow*

$$\mathrm{d}\tilde{S}_t = \tilde{S}_t \left((\mu_2 - \nu_2)\mathrm{d}t + \sigma_2^\top \mathrm{d}W_t \right), \qquad \tilde{S}_0 = \frac{KS_0}{F_0}.$$

4.2 The Power Utility Case

In this subsection, we consider the case of a power utility U given by

$$U(x) = \frac{1}{\alpha} x^\alpha, \qquad \alpha \in (-\infty, 0) \cup (0, 1). \tag{4.5}$$

Notice that the effective domain of the power utility in (4.5) is $\mathbb{R}_+$. Hence, we impose an extra solvency constraint $X_t^u \geq 0$ almost surely for all $t \in [0, T]$ and $u \in \mathcal{A}(x)$. Let τ^u denote the bankruptcy time of process $X = X^u$, i.e., $\tau^u := \inf\{t > 0 : X_t^u \leq 0\}$. A standard treatment is to require $X_t^u \equiv 0$ for all $t \geq \tau^u$ (see Sotomayor and Cadenillas (2009)). We still use the notation $\mathcal{A}(x)$ (or $\mathcal{A}(t, x, z)$) to denote the set of admissible strategies when the solvency constraint is present. We summary the findings on the power utility case in Theorem 4.3.

Theorem 4.3. *Suppose* U *is given by* (4.5). *The optimal trading strategy* $u^* = (\theta^*, N^*)$ *to Problem* (2.9) *is given by*

$$\theta_t^* = \frac{\mathcal{C}_1}{1-\alpha} \cdot X_t^* \qquad \textit{and} \qquad N_t^* = \frac{\mathcal{C}_2}{(1-\alpha)KZ_t} \cdot X_t^*, \qquad \forall\, t \in [0, T], \tag{4.6}$$

where constants $\mathcal{C}_1$ *and* $\mathcal{C}_2$ *are defined in* (4.3).

Proof. We delay the proof to Appendix C. □

In the setup, we choose both spot and futures strategies, θ and N, to be the *amount* of \$USD and number of futures contracts. Under the exponential utility, such a choice leads to optimal investment strategy θ^* being a fixed constant. According to Theorem 4.3, in the power utility case, the optimal *proportion* invested in the risky asset is a constant (see also Merton (1969, 1971)). To

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account for this fact, we introduce the proportional version of u, denoted by $\tilde{u} = (\pi, \psi)$, where

$$\begin{cases} \pi_t = \dfrac{\theta_t}{X_t} & \text{and} \quad \psi_t = \dfrac{KN_t}{X_t}, \quad \text{for } t \in [0, \tau^u), \\ \pi_t = c_1, & \text{and} \quad \psi_t = c_2, \quad \text{for } t \in [\tau^u, T], \end{cases} \tag{4.7}$$

Here, c_1 and c_2 are two constants in the admissible region. We can interpret ψ as the ratio of the total USD amount of inverse contracts to the investor's wealth. Using (4.6), we obtain the optimal trading strategy (π^*, ψ^*) by

$$\pi_t^* = \frac{\mathcal{C}_1}{1-\alpha} \quad \text{and} \quad \psi_t^* = \frac{\mathcal{C}_2}{(1-\alpha)Z_t}, \quad \forall\, t \in [0, T], \tag{4.8}$$

With U given by (4.5), the *relative risk aversion* (RRA) is $(1-\alpha)$. The optimality condition (4.8) says that both π^* and ψ^* are still inversely proportional to the RRA. All the discussions from the previous exponential utility case apply in a parallel way to the power utility case here, with amount changing to proportion.

5 Empirical Studies

5.1 Data and Calibration

To conduct empirical studies, we consider a variant of the market model (2.1)-(2.2) with dimension $n = 2$ as follows:

$$\begin{aligned} \mathrm{d}S_t &= S_t \left(\mu_1 \,\mathrm{d}t + \sigma_S \,\mathrm{d}\bar{W}_{1,t}\right), \\ \mathrm{d}F_t &= F_t \left(\mu_2 \,\mathrm{d}t + \sigma_F(\rho \,\mathrm{d}\bar{W}_{1,t} + \sqrt{1-\rho^2}\,\mathrm{d}\bar{W}_{2,t})\right), \end{aligned} \tag{5.1}$$

where $\bar{W}_1$ and $\bar{W}_2$ are two independent Brownian motions. The parameters μ_i $(i = 1, 2)$ and σ_i $(i = S, F)$ are the appreciate rates and volatilities of the spot S and futures F, respectively, and $\rho \in [-1, 1]$ measures the correlation between the price processes S and F. The equivalence between

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these two models is obvious, e.g., $\sigma_S = \sqrt{\sigma_{11}^2 + \sigma_{12}^2} = \sqrt{\nu_1}$ is indeed the volatility of (2.1), and

$$\bar{W}_1 = \frac{\sigma_{11}}{\sqrt{\sigma_{11}^2 + \sigma_{12}^2}} W_1 + \frac{\sigma_{12}}{\sqrt{\sigma_{11}^2 + \sigma_{12}^2}} W_2$$

is a standard Brownian motion under the same probability space.

Since BitMEX is the largest Bitcoin futures trading platform (see Table A.2), we use the trading data from BitMEX in the empirical studies. We access Bitcoin spot and futures prices data from BitMEX exchange through its provided API (application programming interface). In contrast to traditional regulated exchanges (e.g., CME and CBOE), BitMEX operates 24 hours a day and 7 days a week without trading intermission. Currently, only semi-annual and perpetual futures contracts are listed on BitMEX, although quarterly futures contracts were available in the past. On BitMEX, the futures contracts settle on the BXBT30M index, which is a 30-minute TWAP (time-weighted average price) of the BXBT index ending at 12:00 UTC. The BXBT index is derived from the prices of three Bitcoin exchanges (Bitstamp, Coinbase, and Kraken) using equal weight (33.33%), and the composition is subject to changes due to unprecedented exchange instability. For our purpose, two semi-annual futures contracts (face value $K = 1$) are used in numerical studies, initiated on 2018/12/17 and 2019/3/15, respectively.

We use the first futures contract, spanning days from 2018/12/17 to 2019/3/14, as the in-sample data to calibrate model parameters in (5.1). Since BitMEX is an all-day exchange, we use 365 days per year to estimate annual volatility and return. We obtain

$$\mu_1 = 40.85\%,\ \mu_2 = 60.02\%,\ \sigma_S = 59.94\%,\ \sigma_F = 47.00\%,\ \rho = 0.59. \tag{5.2}$$

We use the second futures contract as the out-of-sample data, which span days from 2019/3/15 to 2019/7/4 to test the performance of optimal trading strategies obtained under exponential and power utility functions.

We report summary statistics of in-sample and out-of-sample spot and futures daily returns in Table 1. The results show slight negative (positive) skewness for in-sample (out-of-sample) return. Both spot and futures prices are very volatile, in-sample daily volatility about 3% (annual volatility,

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57%) and out-of-sample daily volatility 4% (annual volatility, 76%). Daily spot returns range from -9.87% to 10.09% and futures returns from -8.32% to 7.44% for in-sample. For out-of-sample data, daily spot returns range from -14.76% to 16.95% and futures returns from -12.17% to 15.18%. Figure 1 plots the Bitcoin spot and futures prices of in-sample and out-of-sample data. In the upward market, from 2019/3/15 to 2019/7/4, the Bitcoin spot and futures prices display higher consistent co-movement with correlation 99.62%.

Table 1: In-sample and Out-of-sample Summary Statistics of Daily Returns

	In-sample 2018/12/17 - 2019/3/14		Out-of-sample 2019/3/15 - 2019/7/4	
Variables	Spot return	Futures return	Spot return	Futures return
Min	-9.87%	-8.32%	-14.76%	-12.17%
25%-percentile	-0.84%	-0.65%	-0.64%	-0.76%
Median	0.06%	-0.05%	0.78%	0.58%
75%-percentile	1.03%	1.02%	2.58%	2.63%
Max	10.09%	7.44%	16.95%	15.18%
Skewness	-0.08	-0.11	0.34	0.50
Kurtosis	5.62	5.47	5.95	5.25
Daily Volatilty	3.14%	2.46%	4.37%	4.23%
Count	87	87	111	111

5.2 Economic Analysis

We compute optimal trading strategy θ^* in Bitcoin spot and N^* in inverse futures for exponential utility in the middle panels 3-4 of Figure 2 and for power utility in the bottom panels 5-6 of Figure 2. In panels 1-2 of Figure 2, we plot the optimal wealth process X^* for different risk aversion levels. Results show that investors with higher risk aversion (larger γ or small α) achieve lower wealth level in both exponential and power utility cases. Furthermore, investors with higher risk aversion invest less amount in both Bitcoin spot and futures.

We obtain optimal trading strategies under the criterion of utility maximization, which is not well adopted by the industry as a performance measure for portfolios. As a result, to study the performance of obtained optimal strategies, we use well accepted Sharpe ratio and its variant Sortino

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ratio.[7] As a comparison, we also consider a benchmark strategy, called "*All-in-spot*" or long-only in spot, which invests 100% in Bitcoin spot and 0% in the risk-free asset and Bitcoin futures. We report the result in Table 2. All strategies perform extraordinarily well (all Sharp ratios above 4.8 and Sortino ratios above 1.8), which is not surprising at all, since the Bitcoin price nearly tripled from 2019/3/15 to 2019/7/4 (see Figure 1). In all cases, we see that optimal strategies outperform the All-in-spot strategy.

Table 2: Sharp Ratio and Sortino Ratio of Optimal Trading Strategies

Utility	γ, α	Sharpe Ratio	Sortino Ratio
Exponential	$\gamma = 0.5$	5.07	2.28
	$\gamma = 1.0$	4.96	2.09
	$\gamma = 1.5$	4.86	1.99
Power	$\alpha = -2$	4.86	1.87
	$\alpha = -0.5$	4.91	1.89
	$\alpha = 0.1$	4.93	1.90
All-in-spot		4.40	1.59

Note. We choose the initial wealth $X_0 = 4$, the risk-free rate 4% and the user-specified target return 4% for calculating Sortino Ratio.

As noted in Table 1, both Bitcoin spot and futures have high volatility, and the correlation between these two assets can be high as well (see lower plot in Figure 1). In the remaining part, we conduct sensitivity analysis on the impacts of ρ (correlation coefficient) and σ_S (spot volatility) on the optimal wealth process X^* and optimal trading strategies u^*. We report the results in Figures 3-4, respectively. We summarize our main findings below.

- The optimal wealth process X^* under exponential utility is a decreasing function of the correlation coefficient ρ, while in the case of power utility X^* is insensitive to ρ. If the correlation between the Bitcoin spot and futures increases, investors should reduce long positions in Bitcoin spot and simultaneously long more Bitcoin inverse futures.

- The impact of the spot volatility σ_S on the optimal wealth process X^* is unclear, we do not find any monotone relation between X^* and σ_S. When it comes to optimal strategies,

[7] The Sortino ratio is a modification of the Sharpe ratio but penalizes only those returns falling below a user-specified target or required rate of return, while the Sharpe ratio penalizes both upside and downside volatility equally, see https://en.wikipedia.org/wiki/Sortino_ratio.

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we observe that, as σ_S increases, investors should allocate less amount in Bitcoin spot and acquire more contracts in Bitcoin inverse futures.

6 Conclusion

We introduce a financial market where both Bitcoin spot and inverse futures are available risky assets to an investor who acts to maximize her expected utility of terminal wealth. We obtain optimal trading strategies in explicit forms under exponential and power utility functions, and use them to conduct empirical studies. We find that optimal strategies deliver excellent Sharpe ratio and Sortino ratio, and outperform a simply long-only strategy in Bitcoin spot. The results from sensitivity analysis show that, when Bitcoin spot becomes more volatile or the correlation between spot and futures increases, investors should long less Bitcoin spot but more inverse futures.

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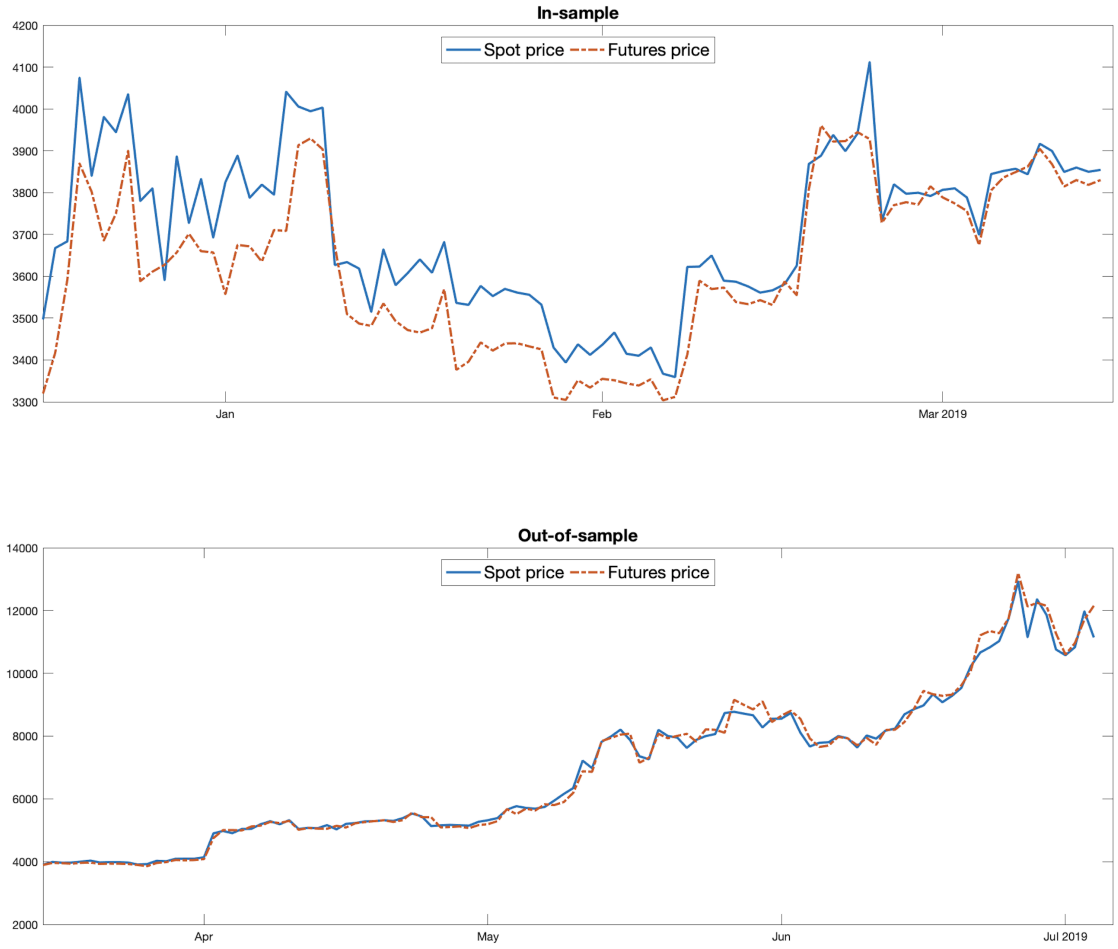
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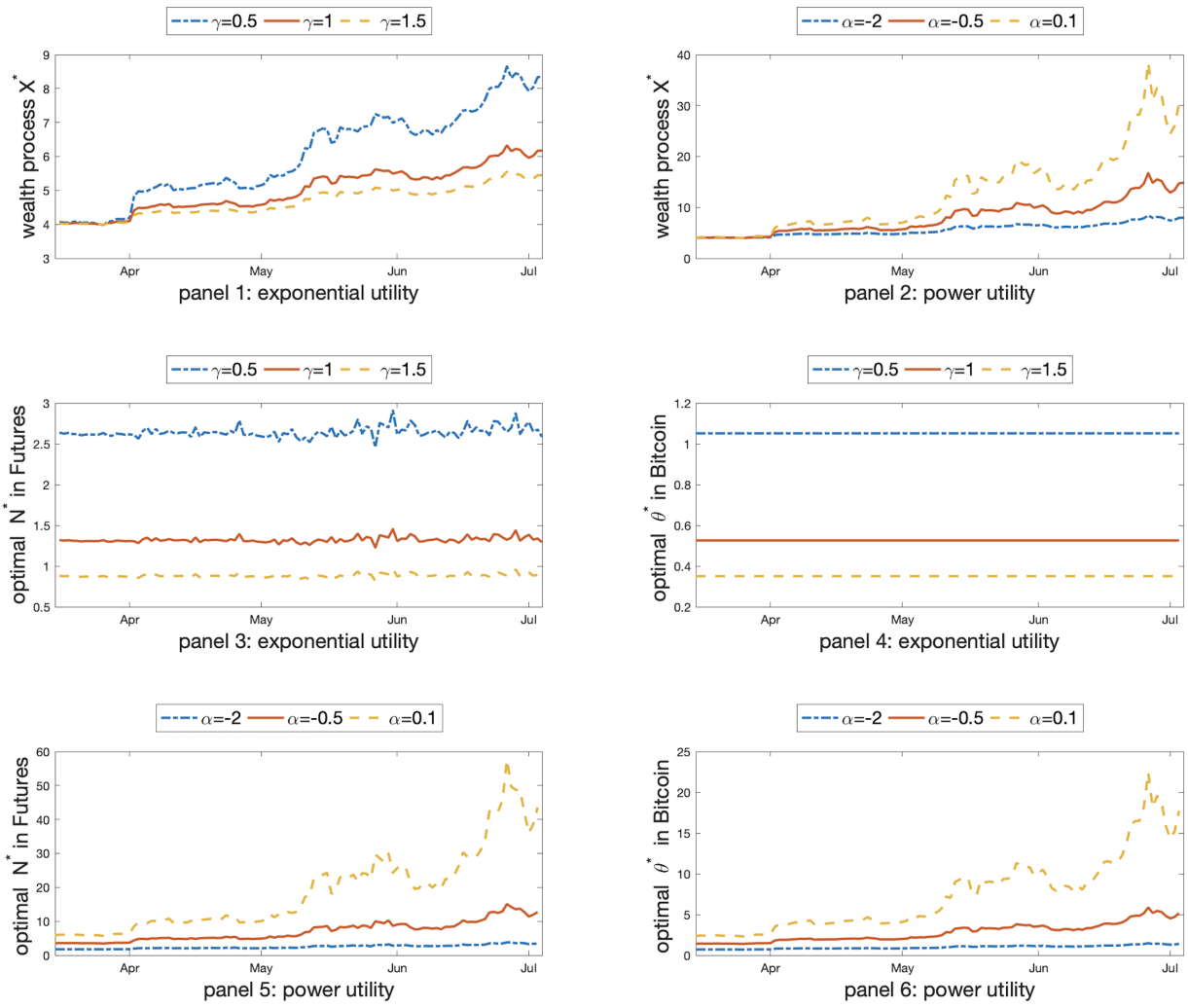
Figure 1: In-sample and out-of-sample Bitcoin Spot and Futures Prices



Note. In-sample dates are from 2018/12/17 to 2019/3/14 and out-of-sample dates are from 2019/3/15 to 2019/7/4.

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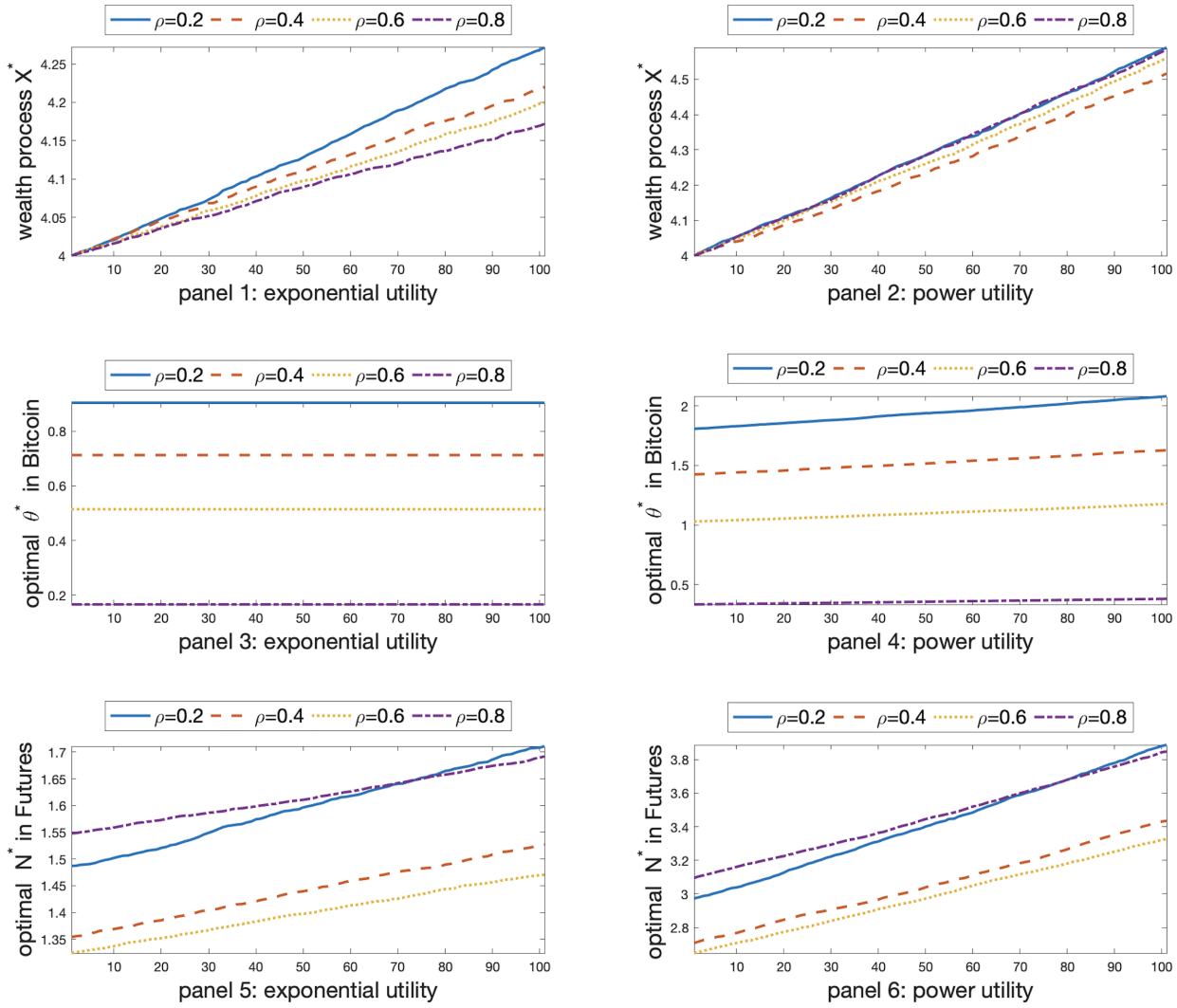
Figure 2: Optimal Wealth Process and Optimal Trading Strategies



Notes. We use model parameters from (5.2) and choose the initial wealth $X_0 = 4$. Panels 1 and 2 report optimal wealth process X^* for exponential and power utility functions. Panels 3 and 4 (resp. 5 and 6) show optimal amount θ^* in Bitcoin spot and inverse futures N^* under different risk aversion levels for exponential (resp. power) utility. In all panels, the x-axis is time from 2019/3/15 to 2019/7/4.

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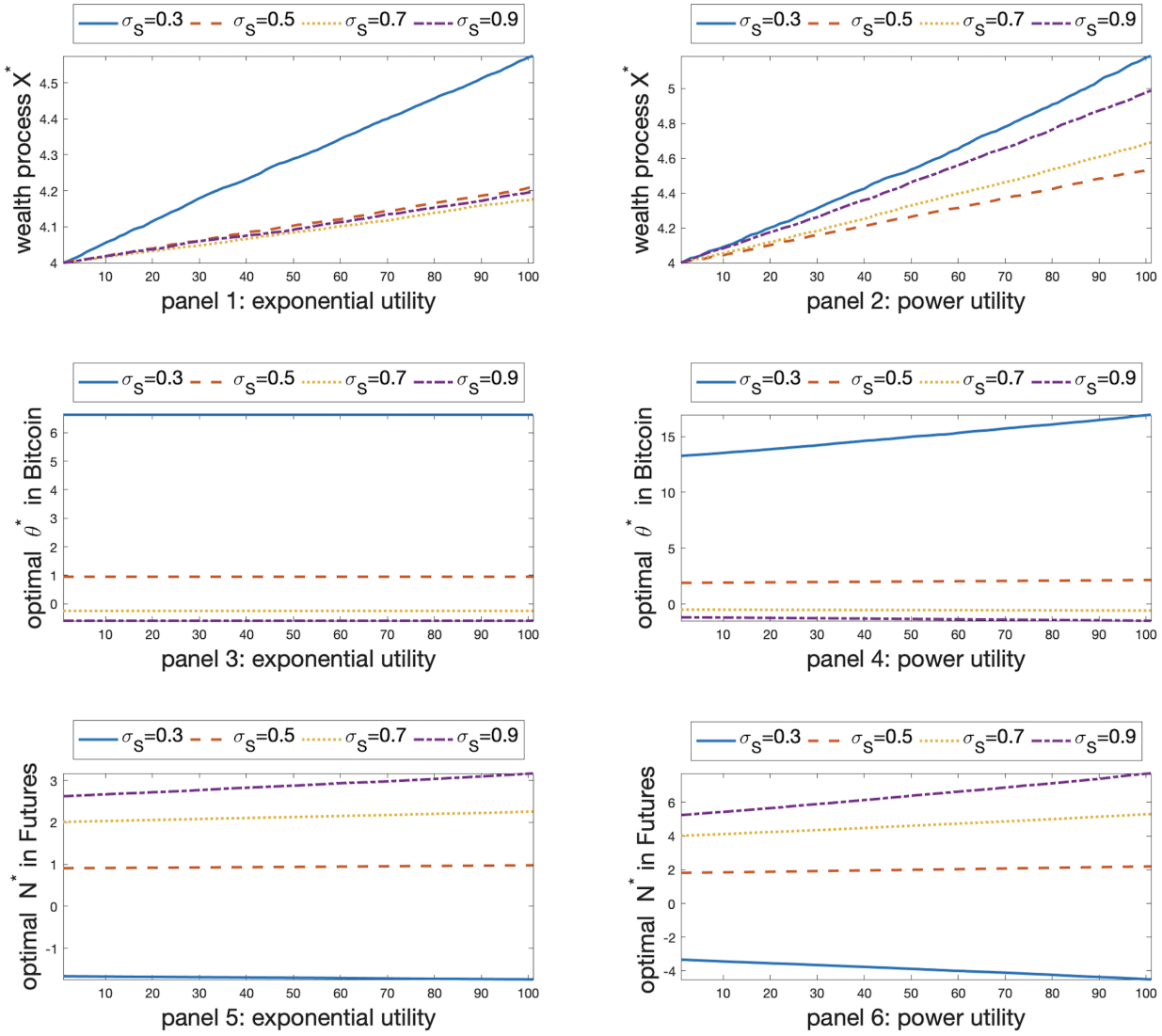
Figure 3: Impact of ρ on Optimal Wealth Process and Optimal Trading Strategies



Notes. We use model parameters from (5.2) (except the correlation coefficient ρ) and choose the initial wealth $X_0 = 4$. Panels 1 and 2 report optimal wealth process X^* under different levels of ρ for exponential and power utility functions. Panels 3 and 4 (resp. 5 and 6) show optimal amount θ^* in Bitcoin spot and inverse futures N^* under different levels of ρ for exponential (resp. power) utility. In all panels, the x-axis is the simulated 100 days.

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Figure 4: Impact of σ_S on Optimal Wealth Process and Optimal Trading Strategies



Notes. We use model parameters from (5.2) (except the spot volatility σ_S) and choose the initial wealth $X_0 = 4$. Panels 1 and 2 report optimal wealth process X^* under different levels of σ_S for exponential and power utility functions. Panels 3 and 4 (resp. 5 and 6) show optimal amount θ^* in Bitcoin spot and inverse futures N^* under different levels of σ_S for exponential (resp. power) utility. In all panels, the x-axis is the simulated 100 days.

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Appendix A Information on Bitcoin and Bitcoin Futures

Table A.1: Ten Largest Cryptocurrencies by Market Capitalization

Name	Symbol	Price	24h Volume ($bn)	Market Capitalization ($bn)
Bitcoin	BTC	$11,881.68	$6.84	$203.17
Ethereum	ETH	$209.65	$2.87	$22.65
Ripple	XRP	$0.29	$0.61	$12.89
Bitcoin-Cash	BCH	$313.39	$0.88	$5.79
Litecoin	LTC	$84.69	$1.40	$5.45
Binance-Coin	BNB	$29.60	$0.19	$4.60
EOS	EOS	$3.91	$1.10	$3.85
Tether	USDT	$0.99	$0.037	$3.57
Bitcoin SV	BSV	$143.41	$0.13	$2.57
Algorand	ALGO	$0.84	$0.088	$2.30

Notes. Currency unit is USD. All the data are accessed at 22:00 CT on August 10, 2019 from www.barchart.com.

Table A.2: Top 5 Exchanges by 24h Bitcoin Futures Trading Volumes

Exchange	24h Volumes ($bn)
BitMEX	2.74
bitFlyer	0.97
Deribit	0.23
CoinFlex	0.17
CryptoFacilities	0.05

Notes. Currency unit is USD. All the data are accessed at 09:06 am UTC on August 20, 2019 from https://www.sk3w.co/bitcoin_futures.

Appendix B Technical Derivations of Methodology

B.1 HJB Characterization

Suppose a smooth value function V exists, and is concave w.r.t. the argument x. By Theorem 3.1, we solve the optimization problem in (3.3) and obtain (the candidate of) an optimal strategy $u^* = (\theta^*, N^*)$ by

$$\theta_s^* = -\frac{\mu_1\nu_2 - \mu_2\nu_3 + \nu_2\nu_3}{\det(\sigma\sigma^\top)}\frac{V_x}{V_{xx}} - Z_t\frac{V_{xz}}{V_{xx}} = -\mathcal{C}_1\frac{V_x}{V_{xx}} - Z_t\frac{V_{xz}}{V_{xx}}, \tag{B.1}$$

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Table A.3: Facts of Standard Bitcoin Futures Contracts at CBOE and CME

Exchange	CBOE	CME
Listing Date	December 10, 2017	December 18, 2017
Terminating Date	NA	June 19, 2019
Ticker	XBT	BTC
Denomination	USD	USD
Contract Size	1 Bitcoin	5 Bitcoins
Margin Rates	40%	35%
Minimum Price Change	\$10 per contract	\$25 per contract
Settlement Reference	Auction Price at the Gemini Exchange	CME CF Bitcoin Reference Rate

Notes. Source: www.investopedia.com and www.cmegroup.com. See also Table 1 of Corbet et al. (2018).

Table A.4: Facts of Bitcoin Inverse Futures Contracts at BitMEX

Ticker	XBT+termination month code
Initial Margin	1.00% + Entry Taker Fee + Exit Taker Fee
Maint. Margin	0.50% + Exit Taker Fee
Risk Limit	50 XBT
Risk Step	50 XBT
Type	Settled in XBT, quoted in USD
Contract Size	1 USD
Settlement	BXBT30M Index Value (12:00:00 PM UTC)
Minimum Price Increment	0.5 USD
Max Price	1000000
Max Order Quantity	10000000
Lot Size	1

Notes. Source: https://www.bitmex.com

$$N_s^* = -\frac{\mu_2\nu_1 - \mu_1\nu_3 + \nu_1\nu_2}{\det(\sigma\sigma^\top)}\,\frac{1}{K\,Z_t}\frac{V_x}{V_{xx}} + \frac{1}{K}\frac{V_{xz}}{V_{xx}} = -\frac{\mathcal{C}_2}{K\,Z_t}\frac{V_x}{V_{xx}} + \frac{1}{K}\frac{V_{xz}}{V_{xx}}, \tag{B.2}$$

where constants ν_i, $i = 1, 2, 3$, are given in (2.4) and $\mathcal{C}_i$, $i = 1, 2$, are defined by (4.3). Both $\mathcal{C}_1$ and $\mathcal{C}_2$ are well defined due to assumption (2.3).

By plugging $u^* = (\theta^*, N^*)$ in (B.1) and (B.2) into the HJB equation (3.2), we obtain

$$V_t + \epsilon_1\left(V_{zz} - \frac{V_{xz}^2}{V_{xx}}\right)z^2 + \left(\epsilon_2\,V_z - \epsilon_3\,\frac{V_xV_{xz}}{V_{xx}}\right)z - \epsilon_4\,\frac{V_x^2}{V_{xx}} = 0, \tag{B.3}$$

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where, to ease notations, we define ϵ_i, $i = 1, 2, 3, 4$, by

$$\begin{aligned}
\epsilon_1 &:= \frac{1}{2}(\sigma_1 - \sigma_2)^\top (\sigma_1 - \sigma_2) > 0, \\
\epsilon_2 &:= \mu_1 - \mu_2 + \nu_2 - \nu_3, \quad \epsilon_3 := \epsilon_2 + \nu_3, \\
\epsilon_4 &:= \frac{((\mu_2 - \nu_2)\sigma_1 - \mu_1\sigma_2)^\top ((\mu_2 - \nu_2)\sigma_1 - \mu_1\sigma_2)}{2\,\det(\sigma\sigma^\top)} > 0.
\end{aligned} \tag{B.4}$$

We consider the following ansatz to the value function of Problem (3.1):

$$V(t, x, z) = f(t, z) \cdot U(x) + g(t, z),$$

where f is a positive function, f and g are smooth enough and satisfy the boundary conditions

$$f(T, z) = 1 \qquad \text{and} \qquad g(T, z) = 0, \qquad \forall\, z > 0. \tag{B.5}$$

Immediately, we have

$$\frac{V_x}{V_{xx}} = \frac{U'(x)}{U''(x)} \qquad \text{and} \qquad \frac{V_{xz}}{V_{xx}} = \frac{f_z(t, z)\, U'(x)}{f(t, z)\, U''(x)},$$

and reorganize the HJB (B.3) into

$$(U f_t + g_t) + \epsilon_1 \left(U f_{zz} + g_{zz} - \frac{f_z^2\, (U')^2}{f\, U''} \right) z^2 + \left(\epsilon_2 (U f_z + g_z) - \epsilon_3 \frac{f_z\, (U')^2}{U''} \right) z - \epsilon_4 f \frac{(U')^2}{U''} = 0,$$

where we have suppressed the arguments for functions f, g and U.

B.2 Martingale Characterization

Suppose there exists a control u^* satisfying the conditions in Lemma 3.2. Since $u^* \in \mathcal{A}(x)$, we have $\mathbb{E}\left[|X_T^*|^2\right] < \infty$ and thus $0 < \mathbb{E}\left[U'(X_T^*)\right] < \infty$. Now we define a new probability measure $\mathbb{Q}$ by

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \frac{U'(X_T^*)}{\mathbb{E}\left[U'(X_T^*)\right]} := H_T$$

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and the Radon–Nikodym derivative process $H = (H_t)_{0\le t\le T}$ by $H_t := \mathbb{E}_t[H_T]$. By definition, H is a $\mathbb{P}$-martingale, and by the martingale representation theorem, has the representation form of

$$\mathrm{d}H_t = -H_t\,\lambda_t^\top\,\mathrm{d}W_t \qquad \text{or} \qquad H_t = H_0\cdot\exp\left(-\int_0^t \lambda_s^\top\,\mathrm{d}W_s - \frac{1}{2}\int_0^t \lambda_s^\top\lambda_s\,\mathrm{d}s\right), \tag{B.6}$$

where $H_0 = 1$ and $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_n)^\top$ is an n-dimensional stochastic process in space $\mathbb{L}^2[0,T]$. Using the Girsanov theorem, we assert that $W^{\mathbb{Q}}$, defined by

$$W_t^{\mathbb{Q}} = W_t + \int_0^t \lambda_s^\top\,\mathrm{d}s, \qquad \forall\, t\in[0,T],$$

is a $\mathbb{Q}$-Brownian motion. By a change of measure and (2.7), we rewrite the condition (3.4) as

$$\mathbb{E}^{\mathbb{Q}}\left[X_T^u\right] = \mathbb{E}^{\mathbb{Q}}\left[\int_0^T \theta_t\left(\mu_1\,\mathrm{d}t + \sigma_1^\top\,\mathrm{d}W_t\right) + KZ_tN_t\left((\mu_2-\nu_2)\,\mathrm{d}t + \sigma_2^\top\mathrm{d}W_t\right)\right] = \text{ constant}, \tag{B.7}$$

for all $u\in\mathcal{A}(x)$, where $\mathbb{E}^{\mathbb{Q}}$ denotes taking expectation under $\mathbb{Q}$.

We consider two special strategies $u_1 = (\theta_1, N_1)$ and $u_2 = (\theta_2, N_2)$, where

$$\theta_{1,t} = x\cdot\mathbb{I}_{t\le\tau}, \quad N_{1,t} = 0, \qquad \text{and} \qquad \theta_{2,t} = 0, \quad N_{2,t} = \frac{1}{K\,Z_t}\cdot\mathbb{I}_{t\le\tau},$$

where $\tau\in[0,T]$ is an arbitrary stopping time and $\mathbb{I}$ is an indicator function. It is clear that both u_1 and u_2 are admissible strategies. By (B.7) and the arbitrariness of stopping time τ, we obtain

$$\left(\sigma_1^\top W_t + \mu_1 t\right)_{0\le t\le T} \qquad \text{and} \qquad \left(\sigma_2^\top W_t + (\mu_2-\nu_2)t\right)_{0\le t\le T}$$

are both $\mathbb{Q}$-martingales. In consequence, this result leads to

$$\sigma_1^\top\lambda_t = \mu_1 \qquad \text{and} \qquad \sigma_2^\top\lambda_t = \mu_2 - \nu_2. \tag{B.8}$$

If the market is complete ($n = 2$), there exists a unique solution to (B.8); otherwise, we have infinitely many choices for λ.

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Given the dynamics of X in (2.7), we apply Itô formula to $U'(X_T^*)$ and get

$$\mathrm{d}U'(X_t^*) = -U'(X_t^*)\left[(\text{diffusion term})\,\mathrm{d}W_t + (\text{drift term})\,\mathrm{d}t\right].$$

By matching the drift term in the above equation with the one in (B.6), we obtain an equality that allows us to express λ using θ^* and N^*. We solve the two equations in (B.8) and find θ^* and N^*.

Appendix C Technical Proofs

Proofs of Theorem 4.1 using the HJB method. With U being an exponential utility, we reduce the HJB into

$$\begin{cases} f_t = \epsilon_4 \cdot f, \quad f_z = 0, \\ g_t = 0, \qquad \epsilon_1 z^2 \cdot g_{zz} + \epsilon_2 z \cdot g_z = 0. \end{cases}$$

Given the boundary conditions (B.5), we obtain the value function to Problem (3.1) by

$$V(t,x,z) = -\frac{1}{\gamma}\, e^{-\epsilon_4(T-t)-\gamma x},$$

which implies u^* in (4.2).

Since $V \in C^{1,2,2}$ is a concave function of x and satisfies the HJB (3.2), V is the value function to Problem (3.1). From (4.2), we observe that both θ^* and ZN^* are constants, hence in $\mathbb{L}^2[t,T]$, and the optimal wealth process X^* is given by

$$X_t^* = x + \frac{1}{\gamma}\big(\mu_1\mathcal{C}_1 + (\mu_2-\nu_2)\mathcal{C}_2\big)t + \frac{1}{\gamma}(\mathcal{C}_1\sigma_1 + \mathcal{C}_2\sigma_2)^\top W_t, \qquad \forall\, t\in[0,T],$$

which together confirm u^* is admissible. The proof is now complete. □

Proofs of Theorem 4.1 using the martingale method. Recall $U'(x) = e^{-\gamma x}$. By applying Itô for-

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mula, we derive

$$\frac{\mathrm{d}U'(X_t^*)}{U'(X_t^*)} = -\gamma\left(\theta_t^*\sigma_1 + KZ_tN_t^*\sigma_2\right)^\top \mathrm{d}W_t - \gamma\left(\mu_1\theta_t^* + (\mu_2-\nu_2)KZ_tN_t^*\right)\mathrm{d}t + \frac{1}{2}\gamma^2\left(\theta_t^*\sigma_1 + KZ_tN_t^*\sigma_2\right)^\top\left(\theta_t^*\sigma_1 + KZ_tN_t^*\sigma_2\right)\mathrm{d}t. \tag{C.1}$$

By matching the diffusion term in (B.6), we obtain

$$\lambda_t = \gamma(\theta_t^*\sigma_1 + KZ_tN_t^*\sigma_2),$$

and, by plugging the above λ_t into (B.8), the system of equations for θ^* and N^* as follows

$$\begin{cases} \nu_1\cdot\theta_t^* + \nu_3 KZ_t\cdot N_t^* = \dfrac{\mu_1}{\gamma}, \\ \nu_3\cdot\theta_t^* + \nu_2 KZ_t\cdot N_t^* = \dfrac{\mu_2-\nu_2}{\gamma}. \end{cases}$$

Due to (2.3), the above system bears a unique solution (θ^*, N^*), which is given by (4.2).

The proof is then complete once we verify u^* is admissible (which is already done in the previous proof) and the two means of computing H_T, (B.6) and (C.1), are consistent. The latter can be done via straightforward but tedious calculus, see proofs in Section 3.2 of Zou and Cadenillas (2014). □

Proof of Theorem 4.3. Let us work with proportional strategies $\tilde{u}$, as defined in (4.7). For any $\tilde{u}\in\mathcal{A}(x)$, the associated wealth process $X = X^{\tilde{u}}$ follows the stochastic differential equation (SDE):

$$\mathrm{d}X_t = X_t\left(\mu_1\pi_t + (\mu_2-\nu_2)Z_t\psi_t\right)\mathrm{d}t + X_t\left(\pi_t\sigma_1 + Z_t\psi_t\sigma_2\right)^\top\mathrm{d}W_t, \qquad X_0 = x > 0.$$

(1) Using the HJB method, we try the ansatz and, after simplifications, obtain $g\equiv 0$ and

$$f_t + \frac{\alpha}{1-\alpha}\epsilon_4\cdot f = 0, \qquad f_z = 0, \qquad f(T,z) = 1.$$

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Therefore, the value function V to Problem (2.9) is given by

$$V(x) = \frac{1}{\alpha}\, e^{\frac{\alpha}{1-\alpha}\epsilon_4 T}\, x^\alpha,$$

where ϵ_4 is defined in (B.4). Plugging V into (B.1) and (B.2) and using the definition of proportional control $\tilde{u}$ yield optimal control (π^*, ψ^*) given in (4.8). The verification step is similar to that of the previous proof.

(2) Under the martingale method, we have

$$\mathrm{d}U'(X_t^*) = -(1-\alpha)U'(X_t^*)\left(\pi_t^*\sigma_1 + Z_t\psi_t^*\sigma_2\right)^\top \mathrm{d}W_t \,+\, (\text{drift term})\,\mathrm{d}t,$$

and, as a result,

$$\lambda_t = (1-\alpha)\left(\pi_t^*\sigma_1 + Z_t\psi_t^*\sigma_2\right).$$

Recall (4.7), we obtain the linear system of (θ^*, N^*) by

$$\begin{cases} \dfrac{\nu_1}{X_t^*}\cdot\theta_t^* + \dfrac{\nu_3 K Z_t}{X_t^*}\cdot N_t^* = \dfrac{\mu_1}{1-\alpha} \\ \dfrac{\nu_3}{X_t^*}\cdot\theta_t^* + \dfrac{\nu_2 K Z_t}{X_t^*}\cdot N_t^* = \dfrac{\mu_2-\nu_2}{1-\alpha} \end{cases}$$

which leads to (4.6). □

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